YEAR 2 CSE C++ CAT 1

TIME ALLOWED: 1 HOUR

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Question 1: (6 Marks)

Write a **function template** named findMax that returns the maximum of two values of any data type (e.g., int, float, char).
Then, write a short main() function that demonstrates the template with **at least two different data types**.

Sample output

Max of 10 and 25 is 25
Max of 4.5 and 2.3 is 4.5

Question 2: 14 Marks

Background:
A point **P** on a Cartesian plane is represented as a pair of coordinates **(x, y)**.
Given two points **P1(x_1, y_1)** and **P2(x_2, y_2)**:

- The **midpoint (P3)** between them is given by:

$x_3 = (x_1 + x_2) / 2$

$y_3 = (y_1 + y_2) / 2$

The **distance (D)** between them is given by:

$D = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$

Tasks:
a. Overload the + **operator** to compute and return the **midpoint** between two Point objects.
b. Write a **member function** to return the **distance** between two Point objects as a float (or double).
c. Demonstrate both in main() by creating two Point objects and showing:

- The midpoint of the two points
- The distance between them

Sample Output:

Enter coordinates of P1: 3, 4

Enter coordinates of P2: 7, 8

Midpoint = (5, 6)

Distance = 5.65685

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